

WEEKLY DIARY

Week Number	Week 2	Submission Date	20 June 2026
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Tasks to be Done

1.	Explore network traffic datasets (CICIDS2017/2018, UNSW-NB15, NSL-KDD) and select the most suitable one for the project
2.	Perform data preprocessing — remove missing values, encode categorical features, normalise numerical values
3.	Split the dataset into training and testing sets
4.	Prepare dataset description and cleaned dataset as deliverables

Tasks Completed

1.	selected UNSW-NB15 as it is modern (2015), manageable in size (~46 MB), contains 9 realistic attack categories, and is the most widely used dataset in the federated learning literature reviewed in Week 1
2.	Loaded UNSW-NB15 training and testing sets (175,341 total records, 45 features); explored data types, class distribution and 9 attack categories
3.	Cleaned the dataset — checked for duplicates and missing values (0 found, confirming dataset quality); dropped non-predictive columns (id, attack_cat)
4.	Trained baseline Logistic Regression and Random Forest models on the cleaned centralised data to establish benchmark performance before federated training: Logistic Regression achieved 93.2% accuracy and 95.2% F1-score; Random Forest achieved 96.1% accuracy and 97.1% F1-score

Remarks / Observations

Week 2 tasks completed successfully. UNSW-NB15 selected as the final dataset after comparing it with NSL-KDD and CICIDS2017/2018. Dataset preprocessing pipeline built and verified — data cleaning, encoding, normalisation, and an 80:20 train-test split completed with zero data loss. Training data further split into 4 client partitions to simulate federated learning participants. Baseline Logistic Regression and Random Forest models trained as performance benchmarks (96.1% accuracy for Random Forest). Ready to proceed with Week 3: federated model design using Flower (FedAvg).

Signature of Student

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Signature of Guide

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